

Multi-Country self-supervised Training and Downstream Task Verification for Colorectal Cancer

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Abstract

The high-resolution and wide field-of-view characteristics of whole slide images (WSIs) often make refined annotation impossible. Fortunately, self-supervised learning (SSL) provides an ability to train models on large-scale unlabeled WSIs directly. However, considering the difference of ethnic and diagnostic criteria, the self-supervised model trained over the data of only one country will inevitably face the problem of poor generalization. In this paper, for colorectal cancer (CRC), we demonstrate the superiority and generalizability of multi-country self-supervised training in four CRC-related medical tasks. We build a multi-country

CRC dataset with 2,249 WSIs from Australia and the United States for SSL. The pre-trained model by SSL further verifies the superiority over four multi-country downstream tasks, including fully supervised tissue classification, biomarker prediction, weakly supervised classification and grading, and five-year survival prediction. Our results show that the performance over the multi-country dataset is better than that over the ImageNet pre-trained model in most downstream tasks. Furthermore, compared with the pre-trained model of single-country data, better generalization can be achieved in the external datasets. In the future, our multi-country data training weights for CRC will be open source, which can better promote the level of artificial intelligence aided diagnosis.

Keywords: Multi-country data, Colorectal cancer, Self-supervised learning, Downstream task verification

1 Introduction

Histopathological diagnosis is currently the gold standard in clinical tumor diagnosis and prognosis evaluation (Srinidhi et al, 2021; Baxi et al, 2021). With the advent of pathological scanners, pathological diagnosis has entered the stage of digitization, and the emergence of a large number of whole slide images (WSIs) has prompted the continuous development of artificial intelligence in this field. Works such as (Campanella et al, 2019; Yu et al, 2021; Lu et al, 2021; Saldanha et al, 2022) prove that the models trained over a large number of WSIs can achieve comparable results with human pathologists on a range of common cancers like colorectal cancer (CRC).

Globally, CRC is the second and third most frequently diagnosed cancer in women and men, respectively, and the incidence and mortality of CRC vary widely around the world (Islami et al, 2021; Arnold et al, 2017). At present, deep learning technology has made a series of breakthroughs in tumors and normal classification of CRC (Zhou et al, 2020; Yu et al, 2021), prediction of microsatellite instability (MSI) (Kather et al, 2019b; Pollett, 2020; Yamashita et al, 2021), CRC grading (Shaban et al, 2020), and prognosis analysis (Kather et al, 2019a; Skrede et al, 2020).

Actually, training from scratch needs large scale of labeled data, which is unrealistic in the histopathology scenario due to the huge number of pixels per WSI and professional knowledge required in labeling (Srinidhi et al, 2021). To alleviate this problem, ImageNet weights pre-trained on natural images are often used for model initialization. Alternatively, another solution is the self-supervised learning (SSL) method (Chen et al, 2020; Chen and He, 2021; He et al, 2021; Xie et al, 2021), which can be employed for unlabeled data. The self-supervised method learns the similarity and dissimilarity of images under different views through data augmentation. Current self-supervised methods have exhibited superior ability to significantly facilitate the learning of downstream tasks in pan-cancer (Ciga et al, 2022; Wang et al, 2021), breast cancer

(Chen and Krishnan, 2022; Srinidhi et al, 2022), and lung cancer (Li et al, 2021), etc.

Notably, another inevitable problem in the histopathology scenario is the difference in tissue preparation and image scanning quality in various hospitals, so studies in one country or center may lead to overly optimistic conclusions (Van der Laak et al, 2021; Hashimoto et al, 2020). Since the developed algorithms need robustly adapt to various changes in clinical practice, multi-country research is considered urgent. At present, some studies have adopted multi-country test datasets to verify the superiority and generalization of the algorithm (Naik et al, 2020; Lu et al, 2021; Laleh et al, 2022). However, as one of the most prevalent cancer types worldwide, CRC specified self-supervised work in multi-country has not been discussed in detail.

This study explores SSL on multi-country data for CRC and evaluating on four downstream tasks. At the SSL pre-training stage, we employ the SimSiam training method over multi-country datasets, which include two large-scale CRC datasets: the MCO dataset from Australia and the TCGA dataset from the United States. At the stage of downstream validation, the pre-trained model is further evaluated over four CRC-related tasks involving datasets from multiple countries. To sum up, we prove that multi-country self-supervised pre-trained model can effectively boost performance on downstream validation datasets in various countries and be more robust in the external dataset.

2 Results

2.1 SSL can be used to pre-train AI models for CRC

We design a pipeline that enables pre-training of AI models under SSL over multi-country dataset, and achieve superior performance on four challenging downstream tasks in CRC (Fig. 1). Although ImageNet pre-trained model can be employed to extract features straightforwardly, there are huge morphological differences between pathological images and natural images. Besides, limited by the lack of annotation in pathological images, self-supervised is considered as a promising solution. To verify the superiority and robustness of SSL over multiple countries, we design a SSL pre-training and downstream task verification framework for CRC. Firstly, we build a multi-country CRC dataset from MCO (1611 WSIs from Australia) and TCGA-COAD/READ (637 WSIs from the U.S.) for pre-training. As shown in Fig. 1a, all WSIs are first cut into a series of patches, and then we adopt a ResNet-50 trained on NCT-CRC-HE to build a more informative pre-training dataset from both TCGA and MCO. Moreover, based on the pre-training dataset, as shown in Fig. 1b, the self-supervised method is used to pre-train the AI model. Specifically, we utilize SimSiam to pre-train ResNet-18. Finally, as shown in Fig. 1c, the pre-trained model is evaluated on four multi-country downstream CRC tasks (Table 1), including fully supervised tissue classification, MSI vs. MSS classification, weakly supervised polyps types classification, adenoma grade classification and CRC grading, and five-year survival prediction.

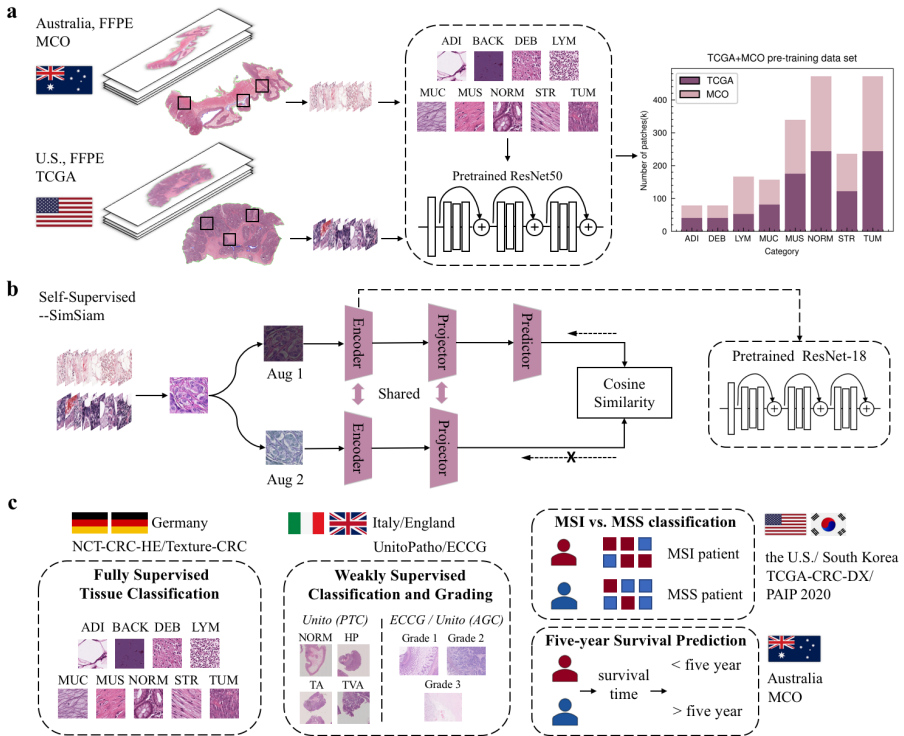


Fig. 1 Schematic of the deep learning and SSL workflows for CRC analysis. **a**, Multi-country dataset construction, a ResNet-50 is trained in NCT-CRC-HE dataset as a tissue classifier to select various patches from the MCO dataset (Australia) and TCGA dataset (the U.S.). Then TCGA pre-training dataset, MCO pre-training dataset, and TCGA+MCO pre-training dataset are used for self-supervised training. **b**, Models employed for self-supervised training, the pre-training method SimSiam is used to obtain the pre-trained ResNet-18 model. **c**, Downstream tasks validation, including fully supervised tissue classification, patient-level classification of microsatellite instability (MSI) vs. microsatellite stability (MSS), weakly supervised classification and grading, and five-year survival prediction.

2.2 SSL can improve fully supervised tissue classification for CRC

We employ SimSiam to pre-train the ResNet-18 model and then fine-tune the pre-trained encoder for a downstream fully supervised tissue classification task. To compare ImageNet initialization and SSL initialization of the AI models, we employ two common evaluation protocols, namely fine-tuning and K Nearest Neighbor algorithm (KNN), to evaluate the performance over the NCT-CRC-HE dataset (overall 107,800 patches from Germany; 9,000 patches are used for training and 7180 patches are used for testing, more details are presented in Sec. ??). Meanwhile, we report the performance over the Texture-CRC dataset which is used as an external testset (overall 5,000 patches from Germany; 4,275 patches are used for testing) under fine-tuning protocol.

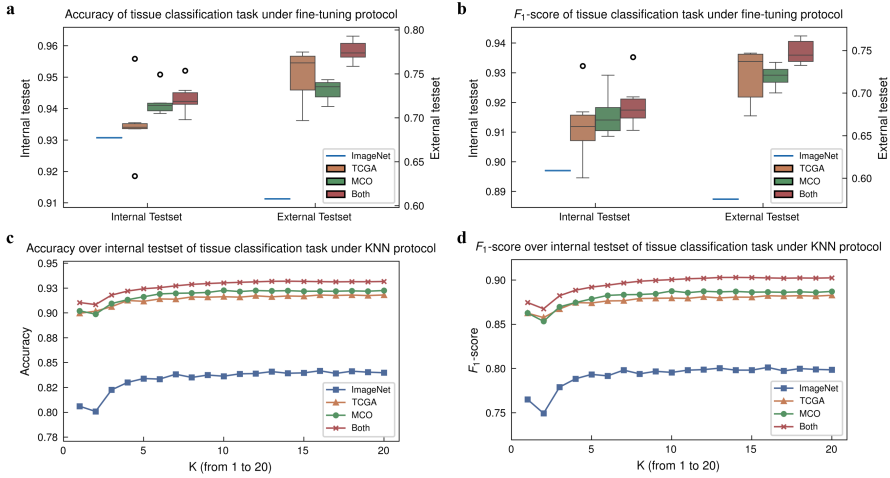


Fig. 2 Results of CRC tissue classification for ResNet-18 (pre-trained by SimSiam). **a**, Classification accuracy over internal and external testset under fine-tuning evaluation protocol. The internal testset refers to NCT-CRC-HE testset and the external testset refers to Texture-CRC dataset. **b**, Classification F_1 -score over internal and external testset under fine-tuning evaluation protocol. In a and b, the boxes show the median values and quartiles, the whiskers show the rest of the distribution (except for points identified as outliers which have a distance of 1.5 times the IQR from quartiles, where IQR is the distance between the upper and lower quartiles). **c**, Classification accuracy over internal testset under KNN evaluation protocol. **d**, Classification F_1 -score over internal testset under KNN evaluation protocol. In c and d, multiple KNN classifiers with K varying from 1 to 20 are constructed and the model performance curves with K are plotted.

Under fine-tuning evaluation protocol, we find that ResNet-18 achieves ACC/ F_1 of 0.9339/0.9044 over the internal testset (NCT-CRC-HE) and ACC/ F_1 of 0.6080/0.5756 over the external testset (Texture-CRC) when using the ImageNet initialization (Fig. 2 and Supplementary Table 1, 2). In comparison, self-supervised pre-training over large colorectal pathological datasets improves the performance of the AI model. Specifically, ACC/ F_1 is improved by 1.74%/2.36%, 1.88%/2.6% and 1.56%/2.2% over the internal testset when SimSiam is used to pre-train the backbone over TCGA, MCO and Both (TCGA+MCO) datasets, respectively. The improvements over external testset are even much more significant, achieving 14.15%/15%, 12.44%/14.91% and 16.88%/17.76%, respectively (Fig. 2 and Supplementary Table 1, 2). Under KNN (integer K from 1 to 20 are tested) evaluation protocol, we report the performance over the internal testset and find that ResNet-18 achieves best ACC/ F_1 of 0.8418/0.8013 (K=16) when using the ImageNet initialization (Fig. 2). In comparison, self-supervised pre-training over large colorectal pathology datasets improves the performance of ResNet-18 by a large margin. Specifically, best ACC/ F_1 is improved by 7.63% (K=16)/8.14% (K=20), 8.24%/9.34% (K=14) and 8.97%/10.07% (K=20) when SimSiam is used to pre-train the backbone over TCGA, MCO and Both (TCGA+MCO) datasets respectively (Fig. 2).

To sum up, we can observe that compared with fully-supervised pre-training over ImageNet, the AI model can obtain better performance after self-supervised pre-training over large colorectal pathological datasets. In addition, pre-training over multi-country dataset (TCGA+MCO) shows excellent performance compared with pre-training over single country dataset (TCGA or MCO) under both fine-tuning protocol ($\text{ACC}/F_1 +0.81\%/+0.73\%$ against TCGA, $\text{ACC}/F_1 +0.14\%/+0.37\%$ against MCO over internal testset, $\text{ACC}/F_1 +2.73\%/+2.76\%$ against TCGA, $\text{ACC}/F_1 +4.44\%/+2.85\%$ against MCO over external testset) and KNN protocol ($\text{ACC}/F_1 +1.34\%/+1.92\%$ against TCGA, $\text{ACC}/F_1 +0.72\%/+0.73\%$ against MCO over internal testset). The significant improvement over ImageNet initialization over external testset indicates much better generalizing ability of AI model with SSL initialization, which is desired in practical scenes. Another interesting observation is that pre-training over the TCGA dataset leads to better generalization over the external testset than pre-training over the MCO dataset, indicating that pre-training over the dataset consisting of images with more variation can greatly promote model generalization.

2.3 SSL can improve fully supervised biomarker prediction for CRC

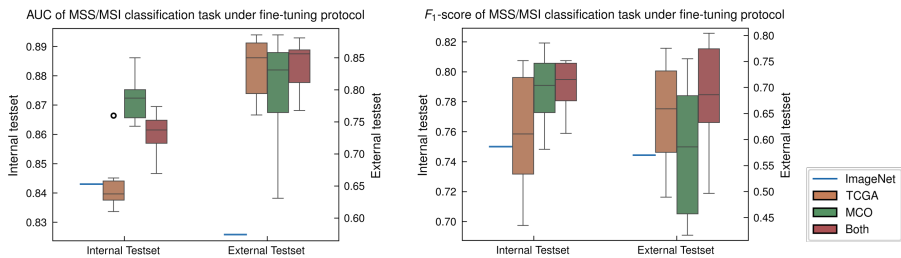


Fig. 3 Results of MSI prediction with ResNet-18 (pre-trained by SimSiam). **a**, Classification AUC over internal and external testset under fine-tuning protocol. The internal testset refers to the TCGA-CRC-DX dataset and the external testset refers to the PAIP2020 dataset. **b**, Classification F_1 -score over internal and external testset under fine-tuning protocol. In a and b, the boxes show the median values and quartiles, the whiskers show the rest of the distribution (except for points identified as outliers which have a distance of 1.5 times the IQR from quartiles, where IQR is the distance between the upper and lower quartiles).

Similar to the process in Sec. 2.2, we transfer the pre-trained backbone of ResNet-18 to another downstream binary classification task (MSI vs. MSS prediction). Considering that the patient-level classification performance highly depends on the patch-level classification performance in this task, and the relative poor performance under KNN protocol may lead to unreasonable patient-level results, we only adopt the fine-tuning protocol to train a patch-level classifier whose predictions for patches are used for acquiring patient-level

prediction. We evaluate the patient-level performance of macro areas under the receiver operating curve (AUC) and F_1 -score for prediction of microsatellite instability over the TCGA-CRC-DX cohort (387 patients from TCGA; more details of this dataset are presented in Sec. ??). We also report the external testing performance over PAIP2020 dataset. Under fine-tuning evaluation protocol, we find that ResNet-18 achieved AUC/ F_1 of 0.8452/0.7376 over the internal testset (TCGA-CRC-DX) and AUC/ F_1 of 0.7881/0.5696 over the external testset (PAIP2020) when using the default ImageNet initialization (Fig. 3 and Supplementary Table 3, 4). Compared with the ImageNet Initialization, self-supervised pre-training over large colorectal histopathology datasets improves the performance of the AI model. Specifically, AUC/ F_1 over internal testset is improved by -0.16%/2.13%, 2.7%/5.02% and 1.49%/5.26%, AUC/ F_1 over external testset is improved by 4.56%/7.82%, 0.99%/0.94% and 4.88%/11.24% when SimSiam is used to pre-train the backbone over TCGA, MCO and Both (TCGA+MCO) datasets, respectively (Fig. 3 and Supplementary Table 3, 4).

To sum up, we can see that the model extracts better features and achieves higher performance on downstream MSI vs. MSS bio-marker prediction task after being pre-trained over large CRC datasets in a self-supervised manner. Meanwhile, pre-training over multi-country dataset (TCGA+MCO) shows excellent performance compared with pre-training over single-country dataset (TCGA or MCO) under fine-tuning protocol (AUC/ F_1 +1.65%/+3.13% against TCGA and F_1 +0.24% against MCO over internal testset, AUC/ F_1 +0.32%/+3.42% against TCGA and AUC/ F_1 +3.89%/+10.3% against MCO over external testset). Moreover, pre-training over TCGA dataset makes the AI model generalize better to the external testset than pre-training over MCO dataset, which is a similar observation in Sec. 2.2.

2.4 SSL can improve weakly supervised classification and grading for CRC

We evaluate weakly supervised CRC classification and grading in the Uni-toPatho cohort (867 WSIs from Italy) and the ECCG cohort (300 WSIs from the UK), and the results are shown in Sec. 2.4 and Supplementary Table 5, 6. In the reported weakly supervised tasks, a WSI often corresponds to only one label. Due to the limitation of computing power, WSIs cannot be directly processed. Therefore, we adopt the general process: 1) splitting small patches, 2) extracting features by a pre-trained model, and 3) training a weakly supervised model. For the ECCG cohort, the self-supervised pre-trained models obtained from pathological images always has better performance than ImageNet. Among the mean metrics of SimSiam training, TCGA (AUC, +0.41%, ACC, +1.52%), MCO (AUC, +0.67%, ACC, +2.17%), TCGA+MCO (AUC, +0.91%, ACC, +2.85%). It can be seen that when using multi-country dataset pre-training over the ECCG dataset, the downstream weakly supervised classification task can achieve better performance. Similarly, we also perform four-category polyps types classification (PTC) and three-category adenoma

grade classification (AGC) over UnitoPatho. Among the mean metrics of SimSiam training, for UnitoPatho PTC, TCGA (AUC, +2.25%, ACC, +3.55%), MCO (AUC, +0.95%, ACC, -0.20%), TCGA+MCO (AUC, +1.69%, ACC, +2.11%), and for UnitoPatho AGC, TCGA (AUC, +1.78%, ACC, +2.65%), MCO (AUC, +3.62%, ACC, +6.85%), TCGA+MCO (AUC, +4.41%, ACC, +6.57%). Overall, it can be seen that the multi-country training still show stable excellence.

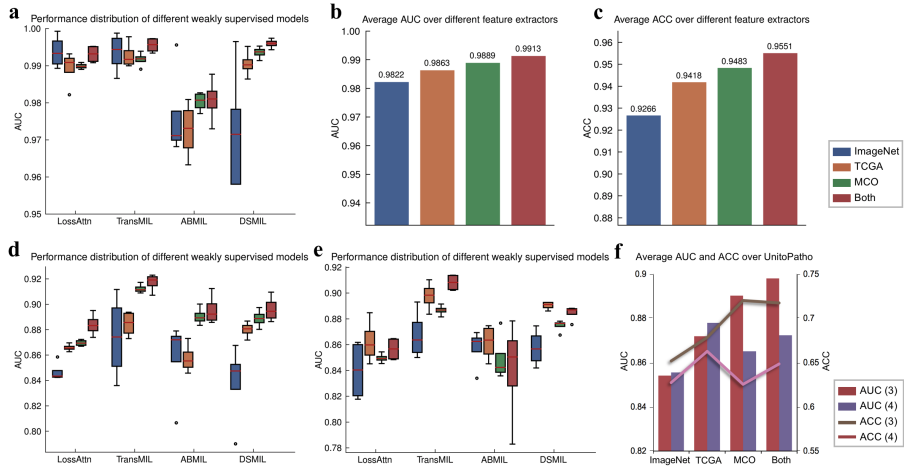


Fig. 4 Results of weakly supervised CRC classification and grading. **a, d, e**, performance distribution for different weakly supervised models over the ECCG grading task, UnitoPatho three-category AGC task and four-category PTC task, respectively. **b, c**, average AUC/ACC with different feature extractors (ImageNet/SimSiam pre-trained) over ECCG grading task. **f**, average AUC and ACC with different feature extractors (ImageNet/SimSiam pre-trained) over UnitoPatho three-category AGC task and four-category PTC task. In the boxplot, the x-axis represents the performance of different weakly supervised models on different pre-training features, and the y-axis represents the AUC interval of the same model on different data partitions. Boxes represent the 1st, median, and 3rd quartiles of the data distribution, with whiskers extending to data points within 1.5 times the interquartile range, and black circles represent outliers. In the bar chart, the four columns represent the average AUC performance of different weakly supervised models on the SimSiam training features. In the combination chart, AUC is shown in the form of a column chart, corresponding to the main coordinate axis. ACC is shown in the form of a line graph, corresponding to the secondary axis.

2.5 SSL can improve weakly supervised five-year survival prediction for CRC

We test our SSL pipeline in another downstream task: CRC five-year survival prediction classification (1611 WSIs from MCO cohort), and the results are shown in Sec. 2.5 and Supplementary Table 7. After removing the data without follow-up time (Details in Sec. 3.4), the MCO cohort retains 1438 WSIs corresponding to 1359 patients. We first cut the WSIs into a series of small patches

and then utilize the pre-trained model to extract features of patches. Finally, we compare the results of four weakly supervised models (Shi et al, 2020; Shao et al, 2021; Ilse et al, 2018; Li et al, 2021) that use pre-trained features on different datasets. In SimSiam pre-training manner, compared to ImageNet pre-trained feature results, the performance gets improved, on average, over TCGA (AUC, +0.26%, ACC, +0.80%), MCO (AUC, +1.59%, ACC, +1.67%), and TCGA+MCO (AUC, +1.35%, ACC, +1.57%). It can be seen from the results that on this task, the pre-training of the MCO dataset and the multi-country dataset can both achieve relatively better performance. This is mainly due to the following two reasons: firstly, the MCO dataset belongs to a specific CRC subtype, that is, the feature distribution of the dataset is relatively simple. Secondly, in this task, we notice that the similarity degree between MCO pre-training dataset and five-year survival prediction downstream dataset is high, which makes the pre-training over MCO dataset has more advantages. Nevertheless, multi-country training still has good adaptability in this special case, which represents higher application value.

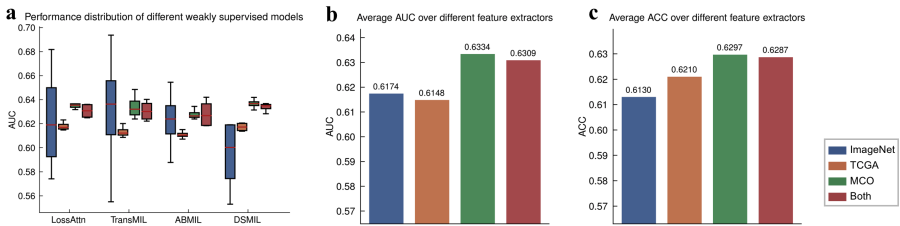


Fig. 5 Results of CRC five-year survival prediction. **a**, results of performance distribution for different weakly supervised models over the MCO dataset. **b**, average AUC and ACC for different features over the MCO dataset. In the boxplot, the x-axis represents the performance of different weakly supervised models on different pre-training features, and the y-axis represents the AUC interval of the same model on different data partitions. Boxes represent the 1st, median, and 3rd quartiles of the data distribution, with whiskers extending to data points within 1.5 times the interquartile range, and black circles represent outliers. In the bar chart, the four columns represent the average AUC performance of different weakly supervised models on the SimSiam training features.

3 Methods

3.1 Ethics statement.

Datasets used in this study have been obtained with appropriate permissions and ethics approvals from the respective data custodians. The TCGA, NCT-CRC-HE, Texture-CRC, TCGA-CRC-DX, PAIA2020, ECCG and UnitoPatho dataset were all publicly available. The MCO CRC dataset ethics was approved by the UNSW Sydney's Research Ethics Committee (UNSW HREAP 2014-7-49 and UNSW HREAP, HC17439). All MCO CRC patients gave written informed consent.

3.2 Patient cohorts.

For CRC patients, we collect five cohorts from five countries in the pre-training and downstream tasks. The MCO cohort from Australia and the TCGA cohort from the United States are used to pre-train the AI model in a self-supervised manner, and the cohorts from other countries are used for both the training and testing in downstream tasks. At the same time, MCO and TCGA cohorts also participate in the downstream tasks, as shown in Table 3.2. We collect cohorts from multiple countries, aiming to represent various clinical conditions in real-world : (1) the MCO cohort (1,611 WSIs, Australia) is a collection of image, clinical data and so on from over 1500 Australian patients who suffered CRC from 1994 to 2010; (2) the TCGA-CRC cohort (638 WSIs, U.S.) is a collection of multi-ethnic WSIs from many research centers, including TCGA-COAD and TCGA-READ, corresponding to colon adenocarcinoma and rectal adenocarcinoma, respectively; (3) the NCT-CRC-HE cohort (107k patches, 224×224 , Germany) is a collection of CRC tissues from the NCT Biobank and the UMM pathology archive; (4) the UnitoPatho cohort (867 patches, $15,855 \times 15,855$, Italy) is a collection released from the University of Turin. And it includes the most relevant patch images extracted from 292 WSIs that are labeled for different cancer grades and tissue types; (5) the ECCG cohort (300 patches, 4548×7548 , England) is a collection released from the Tissue Image Analytics (TIA) Centre which is used for CRC grading.

3.3 Self-supervised methods for pre-training.

To efficiently extract pathological image features, we adopt the state-of-the-art self-supervised pre-training methods SimSiam. SimSiam (Chen and He, 2021) is a contrastive SSL method which learns image representations through Siamese networks. It takes as input x_1 and x_2 which are two randomly augmented views from an image x . An encoder network f consisting of a backbone (e.g., ResNet (He et al, 2016) with the last linear classification layer discarded) and a projection multi-layer perceptron (MLP) (Chen et al, 2020) is used to process x_1 and x_2 . The weights of encoder f are shared when processing the two views x_1 and x_2 . Then, a prediction MLP (Grill et al, 2020) is used to transform the output of one view and match it to the other view by minimizing their negative cosine similarity. Details are in the Appendix.

3.4 Validation setup for downstream task 1&2.

Considering that different epoch numbers of pre-training can affect the downstream task performance and that the AI model appears to have converged after being pre-trained for 20~40 epochs, we take six pre-trained models with different pre-training epochs (i.e., 20, 24, 28, 32, 36 and 40) for downstream task 1&2, and report their average performance. We adopt two common evaluation protocols in our downstream validation experiments, namely fine-tuning and K Nearest Neighbor (KNN) algorithm. Details are in the Appendix.

Table 1 datasets used from multi-center data sources

Data source	Dataset usage	Sample preparation	Population	Number
MCO	Model pre-training Five-year Survival Prediction	FFPE	Australia	1,611 WSIs
TCGA-COADREAD	Model pre-training	FFPE	U.S.	638 WSIs
NCT-CRC-HE	Tissue classification	FFPE	Germany	107K patches
Texture-CRC	Tissue classification (external testing)	FFPE	Germany	5K patches
TCGA-CRC-DX	MSI vs. MSS classification	FFPE	U.S.	192,312 patches
PAIP2020	MSI vs. MSS classification (external testing)	FFPE	Korea	47 WSIs
UNITOPATHO	Weakly-supervised colorectal polyps types classification /adenomas grading	FFPE	Italy	8669 patches (1812×1812)/ 867patches (15,855×15,855)
ECCG	Weakly-supervised cancer grading	FFPE	England	300 patches (4548×7548)

3.5 Validation setup for downstream task 3&4.

Since the image size in task 3&4 is very large (10 million pixels), it's impractical to process image in a supervised manner, so we adopt the weakly supervised method, which is carried out in the following steps: 1) Cut the large image into a series of small patches; 2) Use the pre-trained model to extract features from the small patches; 3) Train and test the performance of a series of weakly supervised models under different feature representations. Details are in the Appendix.

4 Discussion

With the vigorous development of artificial intelligence and people's increasing attention to medical health, computational pathology has become an important research direction for countries all over the world. At the same time, with the continuous upgrading of digital pathology scanning technology, the pathological diagnosis of hospitals worldwide has gradually tended to digital process, resulting in a large amount of data. However, different from natural and medical images such as CT and MRI, the high resolution and wide field of view of pathological images make image annotation a huge challenge. Therefore, pathological images often lack fine local annotation or even have no annotation. The current AI methods in practice usually take the weights pre-trained on the natural image dataset, i.e., ImageNet, to initialize the parameters rather than using pathological images under fully supervised pre-training. Fortunately, self-supervised learning, as a common way of using unlabeled data, can learn the feature of the image itself by designing pretext tasks. Therefore, a self-supervised model based on pathological images is an ideal solution. However, there is significant variance among the data collected in different countries, which is mainly attributed to the following reasons: 1) different physiological characteristics of patients in different countries and regions; 2) the differentiation of pathological slide preparation in different countries and regions; 3) the differentiation of diagnostic criteria in different countries and regions. Therefore, the self-supervised model trained in only one single country still cannot achieve satisfactory results for the generalization of multi-country data. To address this problem, we conduct a series of explorations for multi-country data training and testing of CRC.

As a common malignant cancer worldwide, the research on CRC is of great clinical significance. Typical tasks in CRC today include the analysis of tissue types, the grading of cancer growth patterns, the classification of the polyps malignancy degree, the analysis of biomarker expression, and the study of prognostic treatments. For tissue types classification, complex decomposing tissue into different tissue categories can help doctors better analyze the micro-environment information of tumor (Kather et al, 2019a). The grading task helps doctors judge the tumor's current growth pattern and predict the tumor's growth and spread according to the tumor's degree of malignancy of the tumor (Barbano et al, 2021). The examination of colorectal polyps can help doctors

find precancerous lesions in the intestinal mucosa and help patients avoid and predict invasive cancer by classifying the degree of malignancy and dysplasia of polyps (Barbano et al, 2021). For biomarker analysis, it not only helps to elucidate the pathogenic mechanism of cancer but also helps doctors to design treatment plans in a targeted manner. For example, microsatellite instability determines whether gastrointestinal cancer patients respond particularly well to immunotherapy (Kather et al, 2019b; Echle et al, 2020). Besides, prognostic analysis of colon cancer can help doctors better choose treatment options, and retrospective prognostic studies can explore high-risk factors in cancer prognosis.

For all the downstream tasks of CRC reported in this paper, including tissue classification, biomarker prediction, slide classification and grading, and five-year survival prediction, it can be seen that self-supervised pre-training in multiple countries can generally achieve better results. In the future, our pre-trained model on large-scale CRC dataset can also be more widely applied to other downstream datasets, effectively promoting the level of artificial intelligence medical care.

To intuitively understand and compare the quality of representations extracted by different models, we adopt the uniform manifold approximation and projection (UMAP) technique (McInnes et al, 2018) to reduce the representations dimensionality for visualization. Fig. 6 shows the 2D UMAP projection of patch representations (512 dimensions) extracted by ResNet-18. We can observe that the patch representations extracted by ResNet-18 using SimSiam pre-training initialization tend to form clusters which are more discriminative across different categories, that is, representations of the same category tend to cluster together while representations of different category tend to be more distant. Besides, the clusters appear to be more compact when using the SimSiam Both (lower right) pre-training initialization compared with that of using the SimSiam TCGA/MCO (upper right/lower left) pre-training initialization, as shown in the red and purple rectangular boxes which corresponds to BACK and TUM categories, respectively. The result of UMAP projection indicates that SimSiam pre-training helps the AI model learn patch representations which can better reserve global structure of the original data, and that SimSiam pre-training over multi-country data can further promote the quality of patch representation than over single-country data.

The pre-training datasets used in this paper includes the MCO data set in Australia and the TCGA dataset in the United States. The spatial distribution of the dataset does not cover more datasets such as South America and Africa. Besides, in the downstream task test of this paper, fully supervised classification and weakly supervised classification are tested on four important tasks of CRC. More tasks such as cell detection and gland segmentation require further testing. Furthermore, we found that the construction of multi-country datasets is crucial for the self supervised learning in CRC. In this paper, we propose a method to assist the construction of a multi-country dataset based on an external nine-category dataset of CRC tissues, and to screen patches

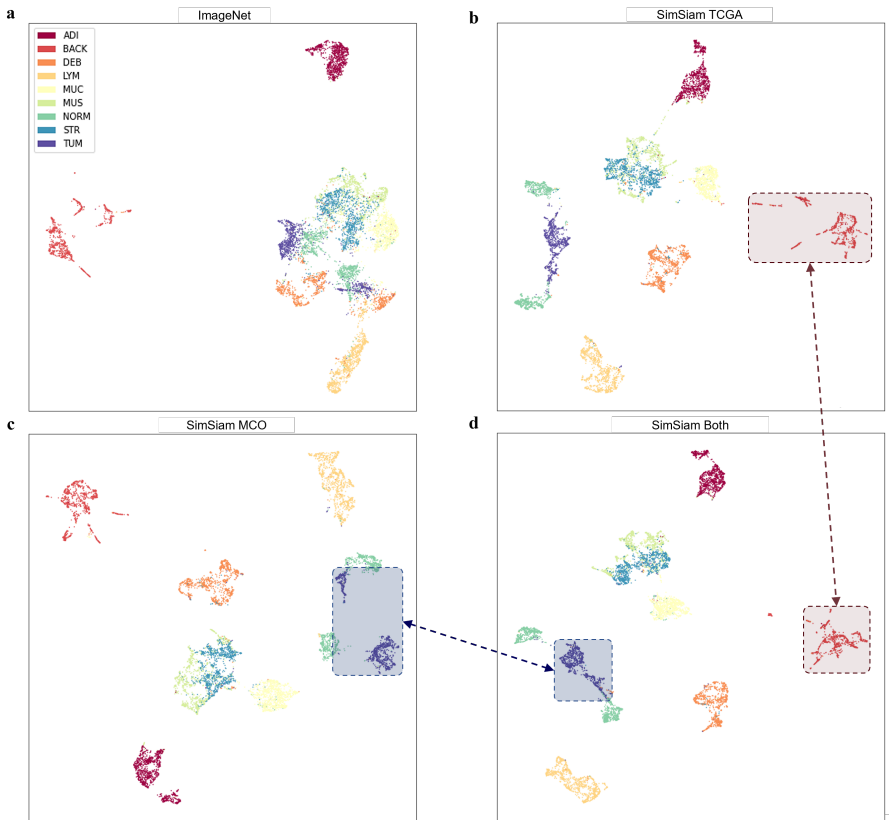


Fig. 6 The 2D UMAP projection of patch representations (NCT-CRC-HE) extracted by ResNet-18 using ImageNet and SimSiam self supervised learning initialization. The 7,180 instances in the testset of NCT-CRC-HE are colored according to their corresponding ground-truth tissue labels by nine different colors listed in the upper left pane. Each instance corresponds to a point in the scatter plot. The instances of BACK in upper right and lower right panes are highlighted by a red rectangular box with dotted line, whilst the instances of TUM in lower left and lower right panes are highlighted by a purple one. We use the default UMAP parameters (neighbors = 15, dist = 0.1) and fix the random state as 42 to further enhance reproducibility.

based on the importance of tissue categories and the complexity of their morphology. The construction method of this multi-country dataset is not ideal enough and deserves to be further improved. In the future, we will build a multi-country CRC dataset covering all continents based on a more sophisticated pre-training set construction mechanism. Furthermore, considering the superior performance of this method in the CRC task, we will apply it to more categories of cancers.

Supplementary information. Experimental results for all downstream tasks are tabulated in Supplementary Tables 1–7.

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Declarations

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- Ethics approval
- Consent to participate
- Consent for publication
- Availability of data and materials
The TCGA dataset is publicly available at the TCGA portal (<https://portal.gdc.cancer.gov>). The MCO dataset is available through the SREDH Consortium (www.sredhconsortium.org), which was used with permission in our current study. The NCT-CRC-HE dataset is publicly available at <https://zenodo.org/record/1214456#.Yx6CbnbP2Uk>. The Texture-CRC dataset is publicly available at <https://zenodo.org/record/53169#.YswA23bP2Um>. The TCGA-CRC-DX dataset is publicly available at <https://zenodo.org/record/2530835#.Ymos2O1ByUk>. Data access for the PAIP2020 can be requested at <https://paip2020.grand-challenge.org/Dataset/>. The ECCG dataset is publicly available at https://warwick.ac.uk/fac/cross_fac/tia/data/extended_crc_grading/. The UnitoPatho dataset is publicly available at <https://zenodo.org/record/4643645#.YOMgBegzaUk>.
- Code availability
- Authors’ contributions

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